COHEN-MACAULAY MORPHISMS OF SCHEMES ARE EXACTLY THE FLAT MORPHISMS WITH COHEN-MACAULAY FIBRES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 5831-5840 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 5842-5844, the complete author proof (1 body lines).

Editorial note. Navigation stubs: 0 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Cohen-Macaulay morphisms and fibres, flatness, and locally Noetherian fibres; the identification is definitional.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of schemes. Assume all fibres of f are locally Noetherian. The following are equivalent*

(1) *f is Cohen-Macaulay, and*
(2) *f is flat and its fibres are Cohen-Macaulay schemes.*

Proof. This follows directly from the definitions. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.